

Data Cleaning Report

Cafe Sales Dataset — Dirty Data Cleaning Exercise

Prepared as portfolio project | Source: Kaggle (ahmedmohamed2003) | Tool: Python / Pandas

Original rows 10,000	Issues found 6 types	Clean rows 9,540	Data retained 95.4%
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1. Dataset Overview

The Cafe Sales dataset is a synthetic dataset of 10,000 point-of-sale transaction records from a cafe. Each row represents a single sale with item details, quantity, pricing, payment method, and location. The dataset was intentionally designed with real-world data quality problems for cleaning practice.

Property	Details
Dataset name	Cafe Sales — Dirty Data for Cleaning Training
Source	Kaggle.com / ahmedmohamed2003
Shape (raw)	10,000 rows x 8 columns
Columns	transaction_id, item, quantity, price_per_unit, total_spent, payment_method, location, transaction_date
All raw data types	All 8 columns stored as object (string) — none were numeric or datetime
Cleaning tool	Python 3 — Pandas, NumPy (Jupyter Notebook)
Output file	cafe_sales_clean.csv — 9,540 rows, fully clean

2. Problems Found (Before Cleaning)

Exploration using `df.isnull().sum()`, `df.duplicated().sum()`, `df.dtypes`, and `df[col].unique()` revealed the following issues:

Issue	Details	Count
Duplicate rows	No exact duplicates found	0 duplicates
Missing values	NaN and 'UNKNOWN'/'ERROR' string placeholders across 7 columns	6,151 total nulls across 7 columns
— item	333 nulls	333
— quantity	138 nulls + string values converted to NaN	479 after type conversion
— price_per_unit	179 nulls + non-numeric strings	533 after type conversion
— total_spent	173 nulls + 'ERROR' string values	502 after type conversion
— payment_method	2,579 nulls	2,579

— location	3,265 nulls	3,265
— transaction_date	159 nulls + unparseable date strings	460 after coerce
Wrong data types	All 8 columns stored as object. Numeric and date columns could not be used for calculations.	8 columns wrong type 8 columns renamed 4 columns affected 1,172 rows incorrect
Column names	Spaces and mixed case in all column headers (e.g. 'Price Per Unit')	
Text inconsistency	Mixed caps, extra whitespace; 'UNKNOWN' and 'ERROR' as invalid category values	
Price mismatch	total_spent did not equal price_per_unit x quantity	

3. Cleaning Steps Applied

Issue	Fix applied	Result
Duplicate rows	df.drop_duplicates() confirmed 0 duplicates	No rows removed
Column names	.str.strip().str.lower().str.replace(' ','_')	8 columns renamed
Date format	pd.to_datetime(errors='coerce') — standardized to datetime64, 460 unparseable rows dropped	460 rows dropped
Numeric types	pd.to_numeric(errors='coerce') on price_per_unit, quantity, total_spent — string errors became NaN	3 columns fixed to float64
Missing values (numeric)	fillna(median) on quantity (479 nulls) and price_per_unit (533 nulls)	1,012 values filled
Missing values (text)	fillna(mode) on item, payment_method, location	6,177 values filled
Text inconsistency	.str.strip().str.title() on all object columns — removed whitespace, standardized case	4 columns cleaned
Price mismatch	Recalculated: total_spent = price_per_unit x quantity as ground truth for all 1,172 mismatched rows	1,172 rows corrected
Remaining total_spent nulls	476 rows where total could not be recalculated (both inputs missing) were dropped	476 rows dropped

4. Before vs After Summary

Metric	Before cleaning	After cleaning
Total rows	10,000	9,540

Duplicate rows	0	0
Missing values (item)	333 nulls	0
Missing values (quantity)	479 nulls	0
Missing values (price_per_unit)	533 nulls	0
Missing values (payment_method)	2,579 nulls	0
Missing values (location)	3,265 nulls	0
Unparseable dates	460	0 — rows dropped
Price mismatches	1,172 rows	0 — recalculated
Column name format	Spaces, mixed case	Lowercase with underscores
Data types	All 8 columns: object	float64 (qty, price, total), datetime64 (date), object (text)
Data retained	—	95.4% (9,540 / 10,000)

5. Cleaning Summary (Notebook)

This is the summary table added as a markdown cell at the end of the Jupyter notebook:

Issue	Details	Fix applied
Duplicate rows	0 found	drop_duplicates()
Missing values	6,151 across 7 columns	Median fill (numeric), mode fill (categorical)
Date format	Mixed formats, 460 unparseable	pd.to_datetime() with coerce
Column names	Spaces, mixed case	Lowercased, underscores
Text inconsistency	Mixed caps, whitespace, UNKNOWN/ERROR values	str.strip() + str.title()
Price mismatch	1,172 rows did not match qty x price	Recalculated total_spent

Final dataset: 9,540 clean rows, ready for analysis.

6. Conclusion

The dataset has been fully cleaned and is ready for analysis. All 6 types of data quality issues have been resolved. The clean output file (cafe_sales_clean.csv) retains 9,540 rows — 95.4% of the original — with zero missing values, correct data types, consistent text formatting, and verified totals.

Notable findings: the dataset had no exact duplicate rows, but suffered from extremely high null rates in location (32.6%) and payment_method (25.8%), plus 1,172 rows where the total_spent field was arithmetically incorrect. All issues were resolved programmatically using pandas.

